

## Research Profile

I develop statistical methodology for heterogeneous, spatially dependent, and censored data, with applications in medicine, environmental health, spatial transcriptomics, and social science.

**Core methods:** Quantile regression; distributed inference; spatial analysis; dimension reduction; survival analysis; missing data analysis; longitudinal data analysis; transfer learning.

## Education

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| 2021–2026 | <b>Ph.D. in Statistics</b> , The George Washington University, Washington, D.C.<br>Dissertation: <i>Quantile Regression for Spatial Data and Censored Data</i> .<br>Advisors: <a href="#">Huixia Judy Wang</a> and <a href="#">Feifang Hu</a> . |
| 2019–2021 | <b>M.S. in Statistics</b> , University of Illinois Urbana-Champaign, IL.                                                                                                                                                                        |
| 2015–2019 | <b>B.S. in Finance</b> , <i>summa cum laude</i> , Kean University, Union, NJ.                                                                                                                                                                   |

## Publications

### Published and Accepted

- [1] **Smoothed Quantile Regression for Spatial Data.**  
Lin, J., Wang, H., Wang, L., and Kim, M. **Accepted**; *Journal of Computational and Graphical Statistics (JCGS)*. Software: [jileil2/SQBiT](#).
- [2] **Hybrid Supervised-unsupervised Modeling for Post-hurricane Private Well Contamination Risk Score Using Empirical Validation and Community-Informed Assessment.**  
Lin, J.<sup>†</sup>, Zhang, J., Wei, E., Shea, K., Wang, H. J., Apanasovich, T. V., Liddie, J. M., Hernandez, E., Norford, C., and Hu, X. C. (2026). *GeoHealth*, 10(6), e2026GH001858. doi:10.1029/2026GH001858.
- [3] **Minimizing Post-shock Forecasting Error through Aggregation of Outside Information.**  
Lin, J. and Eck, D. J. (2021). *International Journal of Forecasting*, 37(4), 1710–1727. doi:10.1016/j.ijforecast.2021.03.010.
- [4] **The Application of PROMETHEE Multi-criteria Decision Aid in Financial Decision Making: Case of Distress Prediction Models Evaluation.**  
Mousavi, M. M. and Lin, J. (2020). *Expert Systems with Applications*. doi:10.1016/j.eswa.2020.113438.
- [5] **Identifying Opinion Leaders in Twitter during Social Events: Social Impact Theory from Influencer’s Perspective.**  
Lin, J., Wang, Z., Huang, Y., and Chen, R. (2020). 2019 4th International Conference on Communication and Information Systems, 197–205. doi:10.1109/ICCIS49662.2019.00042.

<sup>†</sup>Corresponding author.

### Revision-Stage Manuscripts

- [6] **Distributed Inference for Spatial Quantile Regression.**  
Lin, J., Wang, H., and Wang, L. **Reject & Resubmit**; *Journal of the American Statistical Association: Theory and Methods (JASA T&M)*.
- [7] **Transfer Learning for Survival-based Clustering of Predictors with an Application to TP53 Mutation Annotation.**  
Liu, X., Lin, J., Yan, H., Shi, H., Montellier, E., Chi, E. C., Hainaut, P., and Wang, W. **Major revision**; *Journal of the American Statistical Association: Applications and Case Studies (JASA A&CS)*.
- [8] **Censored Bent-line Quantile Regression with Application in a Study on Experimental Autoimmune Myasthenia Gravis.**  
Lin, J., Wang, H., and Luo, J. **Major revision**; *Annals of Applied Statistics (AoAS)*.

### Submitted Manuscripts

- [9] **Striational Antibody-Associated Myositis–Bridging the Gap between Thymoma and Myasthenia Gravis: A Systematic Review.**  
Luo, J., Lin, J., Shymansky, J., and Wang, H. J. **Submitted**; *Nature Communications*. [Code](#).

### Working Papers

- [10] **Adaptive Weighting of Source Datasets for High-Dimensional Transfer Learning in Linear Regression.**  
Lin, J., Wang, H. J., and Liu, M. Work in progress.
- [11] **Graph-Based Transfer Learning for Survival-Based Clustering of Predictors.**  
Lin, J., Parsaee, A., Li, R., Wang, W., and Wang, H. J. Work in progress.
- [12] **3-D Alignment of Tumor Tissue Sections Using Tumor-Specific Total mRNA Expression (TmS).**  
Lin, J. and Wang, W. Work in progress.
- [13] **Cost-Sensitive Ordinal Gradient Boosting with an Application to PFAS Screening in Private Wells.**  
Lin, J., Liddie, J. M., Shea, K., Wei, E., Zhang, X., Tokranov, A. K., and Hu, X. C. Work in progress.

## Software

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| <b>SQBiT</b> | R package for smoothed spatial quantile regression; companion to the JCGS paper. <a href="#">jileil2/SQBiT</a> .                    |
| <b>DISQ</b>  | R package for distributed spatial quantile regression; companion to the JASA T&M revision. <a href="#">jileil2/DISQ</a> .           |
| <b>CBQR</b>  | R package for censored bent-line quantile regression; companion to the AoAS revision. <a href="#">jileil2/cbqr</a> .                |
| <b>PFAS</b>  | Interactive Shiny application for national PFAS prediction and cost-sensitive screening. <a href="#">PFAS Risk Screening Tool</a> . |

## Professional Experience

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| <b>2026–Present</b> | <b>Postdoctoral Associate</b> , The University of Texas MD Anderson Cancer Center.                                                                                                 |
| <b>2025–2026</b>    | <b>Research Assistant</b> , Predicting Drinking Water Contaminants in Underserved Areas with Transfer Learning. Principal investigators: Xindi Hu and Xiaoke Zhang.                |
| <b>2025</b>         | <b>Research Assistant</b> , Data-Driven Solutions to Mitigate the Impact of Hurricane on Private Well, Drinking Water Quality, and Human Health. Principal investigator: Xindi Hu. |
| <b>2024–2025</b>    | <b>Research Assistant</b> , Unlocking Complex Heterogeneity in Large-Scale Spatial-Temporal Data with Adaptive Quantile Learning. Principal investigator: Huixia Judy Wang.        |
| <b>2023–2024</b>    | <b>Research Assistant</b> , Antigen-Specific Immunosuppression of Myasthenia Gravis by CAR-Engineered Tregs. Principal investigator: Jie Luo.                                      |

## Teaching Experience

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| <b>2024</b> | Teaching Assistant, Introduction to Mathematical Statistics II (STAT 4158), Introduction to Business and Economic Statistics (STAT 1051), Mathematical Statistics II (STAT 6202), GWU. |
| <b>2023</b> | Teaching Assistant, Regression Analysis (STAT 2118), Survival Analysis (STAT 6227), Introduction to Business and Economic Statistics (STAT 1051), GWU.                                 |
| <b>2022</b> | Teaching Assistant, Data Analysis (STAT 6210), Mathematical Statistics I (STAT 6201), GWU.                                                                                             |
| <b>2021</b> | Teaching Assistant, Advanced Regression Analysis (STAT 527), University of Illinois Urbana-Champaign.                                                                                  |
| <b>2020</b> | Teaching Assistant, Applied Regression and Design (STAT 425), University of Illinois Urbana-Champaign.                                                                                 |

## Presentations

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| <b>2026</b> | StatConnect 2026.<br><i>Censored Bent-line Quantile Regression with Application in a Study on Experimental Autoimmune Myasthenia Gravis.</i>                                                             |
| <b>2025</b> | The Annual ASA/IMS Spring Research Conference on Statistics in Industry and Technology, New York, NY.<br><i>Distributed Inference for Spatial Quantile Regression.</i>                                   |
| <b>2025</b> | StatConnect 2025, Vienna, VA.<br><i>Distributed Inference for Spatial Quantile Regression.</i>                                                                                                           |
| <b>2025</b> | The Past, Present and Future of Statistics in the Era of AI, Washington, D.C.<br><i>Censored Bent-line Quantile Regression with Application in a Study on Experimental Autoimmune Myasthenia Gravis.</i> |
| <b>2024</b> | ICSA Applied Statistics Symposium, Nashville, TN.<br><i>Distributed Inference for Spatial Quantile Regression.</i>                                                                                       |
| <b>2024</b> | Statistics in the Age of AI, Washington, D.C.<br><i>Censored Bent-line Quantile Regression with Application in a Study on Experimental Autoimmune Myasthenia Gravis.</i>                                 |
| <b>2023</b> | ICSA Applied Statistics Symposium, Ann Arbor, MI.<br><i>Smoothed Quantile Regression for Spatial Data.</i>                                                                                               |
| <b>2018</b> | INFORMS International Conference, Taipei, Taiwan.<br><i>The Application of PROMETHEE Multi-criteria Decision Aid in Financial Decision Making: Case of Distress Prediction Models Evaluation.</i>        |

## Mentoring

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| <b>2024–2025</b> | Maxwell Beveridge, undergraduate student. |
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## Honors and Awards

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| <b>2026</b> | Minna Miran Kullback Memorial Prize for Leadership and Service, Department of Statistics, GWU. |
| <b>2025</b> | Ph.D. Student Poster Competition Award, StatConnect 2025, Vienna, VA.                          |
| <b>2024</b> | GW Statistics Cornfield Travel Award, Department of Statistics, GWU.                           |

## Professional Service

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| <b>Referee</b> | Computational Statistics & Data Analysis. |
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## Skills

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| <b>Programming</b>        | R, Python, Bash, LaTeX.                                                |
| <b>HPC and Automation</b> | SLURM job scripting, GPU/CPU cluster workflows, environment modules.   |
| <b>Reproducibility</b>    | Git/GitHub project management, version control, workflow organization. |